

ATPL EASA STUDY GUIDE | 021 - AIRFRAME & SYSTEMS

Chapter 13: Magnetism, DC Generation & Starter-Generators

Syllabus Source: CAE Oxford (pp. 69-96 (Elec)) | Pass Target: >= 90% AviationExam

1. ELECTROMAGNETIC INDUCTION & DC GENERATOR PRINCIPLES

Electrical generation is based on Faraday's Law of Electromagnetic Induction: when an electrical conductor cuts magnetic lines of flux, an Electromotive Force (EMF / Voltage) is induced in the conductor:

- Fleming's Right-Hand Rule (Generators): Thumb = Motion of conductor; First finger = Magnetic Field (North to South); Second finger = Induced Current direction.
- Fundamental AC Generation in Armature: ALL rotating generators fundamentally produce ALTERNATING CURRENT (AC) inside the spinning armature coils as the wire alternates cutting north and south magnetic poles!
- The Commutator (Mechanical Rectifier): In a DC generator, the rotating armature is connected to a split copper cylinder called a COMMUTATOR. Stationary carbon brushes ride on the commutator segments, mechanically switching connections at the exact instant induced voltage reverses, outputting pulsing DIRECT CURRENT (DC) to the busbar.

2. DC VOLTAGE REGULATION & SHUNT GENERATORS

Aircraft DC generators are self-exciting shunt generators (field coils connected in parallel with the armature). Generated EMF is proportional to: 1. Magnetic field strength (flux); 2. Rotational speed (RPM); 3. Number of armature conductor turns:

- Voltage Control Mechanism: Because engine RPM varies widely across flight regimes (idle to take-off), generator output voltage MUST BE REGULATED by controlling FIELD CURRENT in the stationary field windings.
- Carbon Pile Voltage Regulator: Contains a stack of carbon discs in series with the field winding. An electromagnet sensing bus voltage compresses or releases spring pressure on the carbon stack. Higher voltage -> electromagnet pulls against spring -> decompresses carbon discs -> stack resistance increases -> field current drops -> generator voltage drops back to 28V.
- Reverse Current Circuit Breaker (RCCB): Prevents the battery from discharging backward into the generator when engine RPM drops below generator cut-in speed (motoring the generator). Disconnects generator automatically if current reverses.

[!] EXAM TRAP (AVIATIONEXAM) - Starter-Generator Operating Modes in Turboprops

- Twin-role machine: Acts as a powerful DC electric motor during engine start, then switches automatically to become a 28V DC generator once engine accelerates past self-sustaining starter cut-out speed! - During Start: Series field windings engaged for maximum high-torque starting power. - During Generation: Switches to shunt field windings with voltage regulator active to supply steady 28V DC.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): DC Regulation & Reverse Current

1. DC Voltage Regulation: 28V DC bus output is maintained constant by varying the field winding current (carbon pile or solid-state regulator). 2. Reverse Current Relay (RCR): - When generator RPM drops below idle, its output voltage drops below battery voltage (24V). - RCR opens main line contactor instantly to prevent battery discharging into generator windings.

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the primary operational function of the commutator in a DC aircraft generator?**

- [A]: To increase rotational speed
- [B]: To act as a mechanical rotary rectifier, converting the alternating current (AC) naturally produced in the armature into direct current (DC)
- [C]: To regulate system voltage
- [D]: To insulate the brushes

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. All rotating armatures induce AC. The commutator is a split copper ring that mechanically reverses contact polarity twice per revolution, rectifying AC into pulsating DC.

Q2: How is the output voltage of an aircraft DC generator regulated as engine RPM changes in flight?

- [A]: By altering the number of windings on the armature
- [B]: By varying the strength of the magnetic field current
- [C]: By moving the carbon brushes mechanically
- [D]: By changing battery electrolyte levels

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Output voltage = $k \times \text{Flux} \times \text{RPM}$. Because RPM varies with engine throttle, the voltage regulator modulates the current flowing through the shunt field coils to keep bus voltage constant at 28V.

Q3: What device prevents an aircraft battery from discharging back into a DC generator when engine RPM drops?

- [A]: A carbon pile
- [B]: A Reverse Current Circuit Breaker (RCCB) or reverse current diode
- [C]: A static inverter
- [D]: A transformer rectifier unit

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. If generator voltage drops below battery voltage, current would reverse and try to drive the generator as an electric motor. The RCCB detects reverse current flow and instantly opens the contactor.